

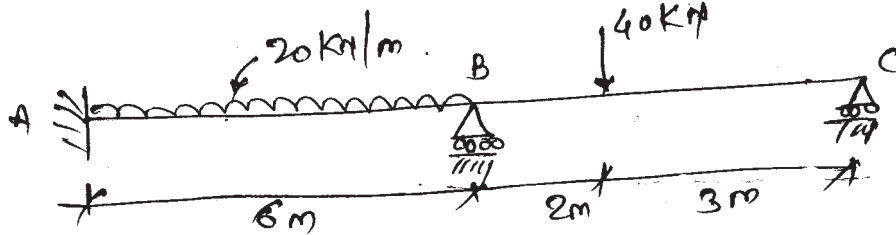
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T.E. (Civil)

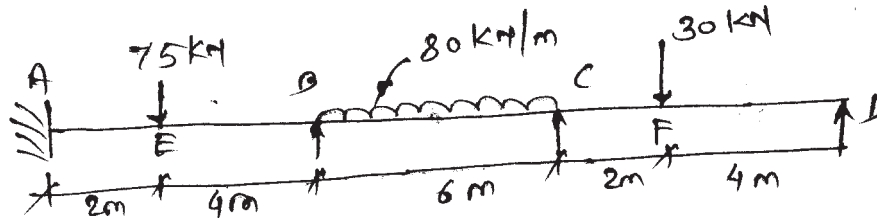
STRUCTURAL ANALYSIS - II**(2012 Course) (Semester - I)***Time : 2½ Hours]**[Max. Marks :70**Instructions to the candidates:*

- 1) Answer questions Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right side indicate full marks.
- 3) If necessary, assume suitable data & indicate clearly.
- 4) Use of electronic pocket calculator is allowed.

- Q1) a)** Analyse the beam by slope deflection method. Draw BMD & SFD. Take $EI = 3900 \text{ kN-m}^2$. The support 'B' sinks by 30mm. **[10]**



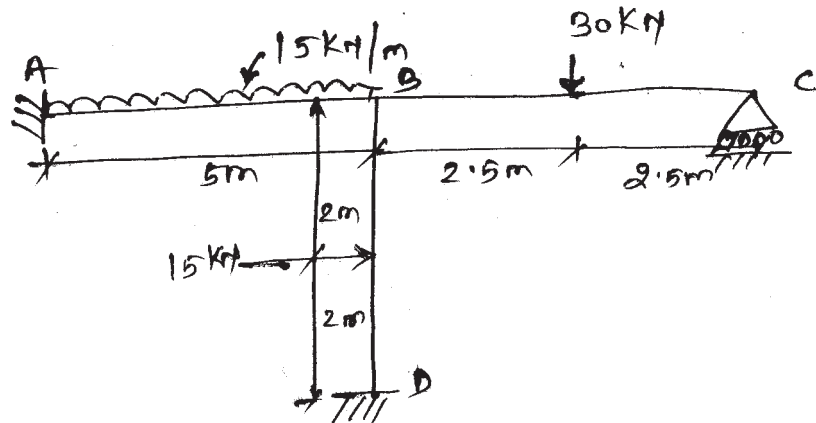
- b)** Analyse the continuous beam shown in figure using flexibility method & draw the bending moment diagramme. **[10]**



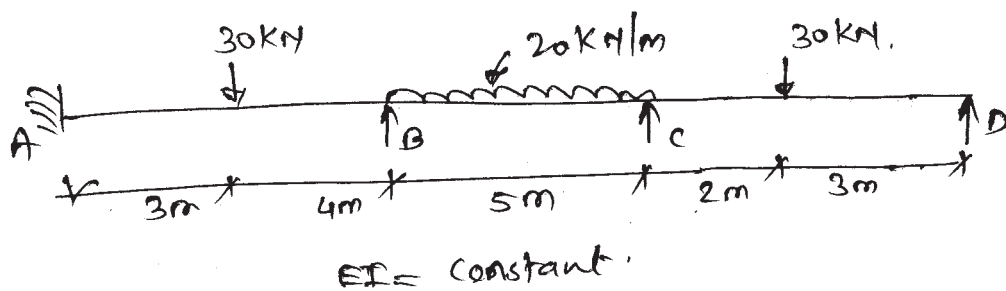
OR

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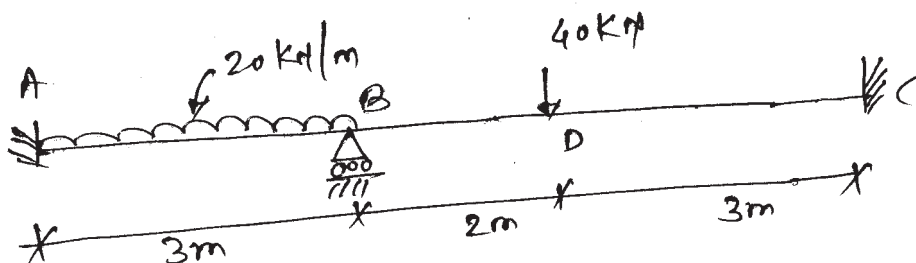
- Q2) a) Analyse the frame as shown in figure. EI is constant. Use slope deflection method. [10]



- b) Analyse the continuous beam shown in figure by moment distribution method. Draw BMD & SFD. [10]



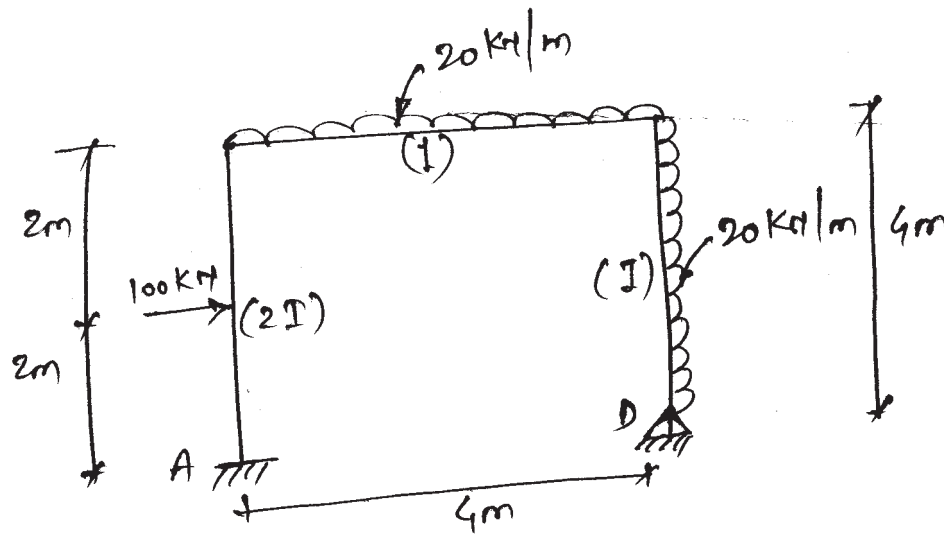
- Q3) Analyse the beam shown by stiffness matrix method. Draw BMD & elastic curve. EI = constant. [16]



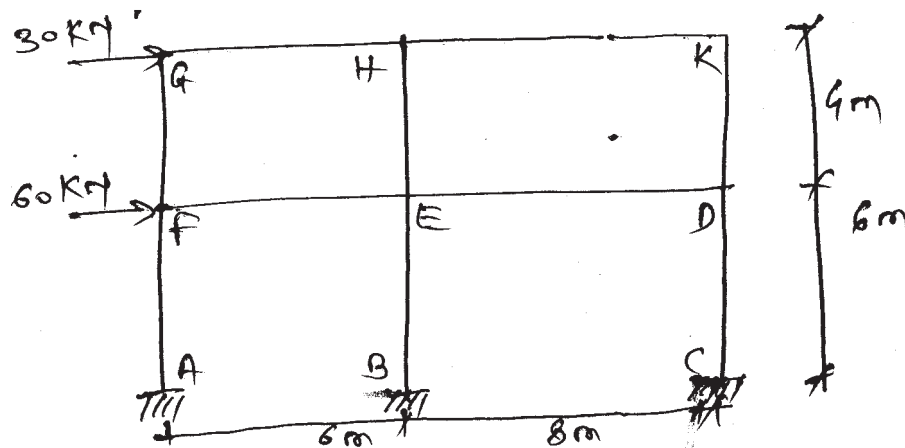
OR

Q4) Analyse the frame by stiffness matrix method & sketch BMD.

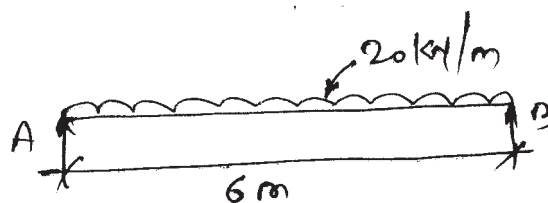
[16]



Q5) a) Determine the approximate values of moment, shear & axial forces in member of the frame loaded & supported as shown in figure using cantilever method of analysis. [12]



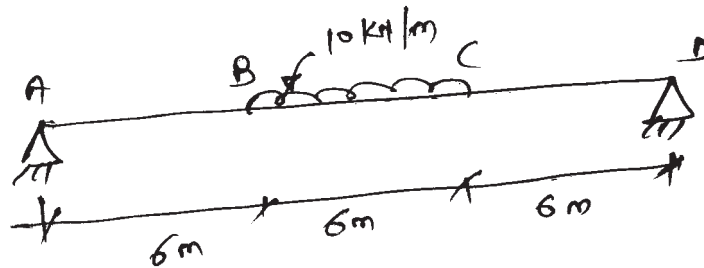
b) A simply supported beam of length 6m is loaded as shown in figure. Determine the maximum deflection. [6]



OR

Q6) a) Analyse the frame as shown in Q.5 (a) by portal method. [10]

b) The beam is loaded & supported as shown in fig. Determine deflection at nodal points. Take 3 nodes. [8]



Q7) a) Explain the terms: [8]

i) Constant strain Triangle.

ii) Linear strain Triangle.

iii) Higher order elements.

iv) Nodes.

b) Explain shape function for Quadratic rectangular element. [8]

OR

Q8) a) Explain shape function & state properties of shape function. [8]

b) Differentiation between Axisymmetric & Isoparametric elements. [8]

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